

RAM / ROM map

Memory maps connect an address bus to stored words

DE AD BE EF starter

- Starter: sixteen-by-eight RAM loaded from a readmemh-style dump
- At address zero: hex DE, then AD, BE, EF
- Read address zero gives data DE
- Read address three gives EF
- The grid shows every cell with at-sign and hex value
- Switch to ROM and try a write, the status reports blocked

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

RAM / ROM address map

Browse a small memory array by address. Write (RAM) or load hex via a \$readmemh-style dump (ROM/init). Cells flash on read and write. Starter: 16×8 RAM with a short hex image.

Starter example: 16×8 RAM loaded with DE AD BE EF at @00 via a \$readmemh-style dump.

Challenge 0 / 22 cleared

Quiz: RAM vs ROM: In this lab, ROM mode means...

☐ runtime writes are rejected; init via \$readmemh-style load

☐ reads are illegal

☐ addresses are decimal only

☐ DEPTH must be 1

Show hintCheckNextIdle

Quiz: RAM vs ROMQuiz: \$readmemhQuiz: @addrQuiz: address bitsQuiz: word vs byteRead addr 0Read addr 3

Write RAMROM blocks writeLoad @08 AAClear memorySelect by cellWrite 0xFFRead after write

Quiz: power-upQuiz: RAMReload starterFill 0..3Quiz: address orderQuiz: DEPTHROM init load

Data bus shows

Memory map starter

Workbook practice

- On paper, draw a sixteen-word memory map with addresses zero through F
- Label starter bytes DE AD BE EF at zero through three
- Explain what at-sign zero A means in a readmemh file
- Tabulate RAM versus ROM: read allowed, write allowed, init method
- Name one pitfall: confusing byte address order with big-endian multi-byte layout elsewhere

Pitfalls to watch

- Do not assume ROM never changes, init load still programs the image in this lab
- At-sign sets address; following bytes fill upward sequentially
- Address bits are log two of depth, sixteen words need four bits, not eight
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, read addr zero and three, then try a ROM-blocked write
- On paper, sketch one readmemh dump with at-sign and four data bytes
- When you are ready, take the short quiz, then continue to FIFO pointers